Introduction to Software Engineering

Project Idea & Team Contract

The student team is required to complete the **Project Idea & Team Contract** documentation for the assigned course project, following the attached template.



Software Engineering Department
Faculty of Information and Technology
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Project Idea & Team Contract

UniEvent – Team T07

Objectives

This document focus on the following topics:

- ✔ Describing the project idea at a high level so the teaching staff can approve it
- ✔ Agreeing on how the team will work together (team contract)
- ✔ PA0 is not graded, but it is required before the team starts PA1

1 Team Information

Item	Value
Team ID	T07
Team name	UniEvent Team
Team lead	Lê Nguyễn Quốc Học (24127044)
Communication channel	Discord server "UniEvent - T07" (https://discord.gg/zVMjhEGnzp); the TA is invited as a member

Student ID	Full name	Email	Main role	Backup role
24127044	Lê Nguyễn Quốc Học	24127044@student.hcmus.edu.vn	Team lead / Project manager; Developer	Business analyst
24127177	Thái Quang Huy	24127177@student.hcmus.edu.vn	Business analyst; Developer	Tester
24127196	Trần Vũ Anh Kiệt	24127196@student.hcmus.edu.vn	Designer / Architect; Developer	Team lead / Project manager
24127205	Nguyễn Văn Minh	24127205@student.hcmus.edu.vn	Tester / DevOps; Developer	Designer / Architect

In this document, A, B, C and D refer to the members in the order of the table above. Our team has four members, so the developer role is shared: every member implements one group of modules, writes its unit tests and reviews other members' pull requests. Every role has one main person and one backup. The final assignment is confirmed in PA1.

2 Project Idea Overview

2.1 Project Name and One-Sentence Pitch

UniEvent - a web platform that lets university student clubs get event approval, sell tickets with automatic VietQR payment confirmation and check attendees in by QR code in seconds, with optional attendance-based conduct score management.

2.2 Problem and Motivation

Student clubs, the Youth Union and the Student Association organise many events every semester: workshops, seminars, competitions, music nights and volunteer activities. In our experience, many of these events are still run with general-purpose tools:

- ✔ **Registration and payment.** Students register with Google Forms and pay by bank transfer; organisers reconcile bank statements against form responses by hand, which is slow and error-prone.
- ✔ **Tickets and check-in.** Confirmations are e-mailed manually; at the door, volunteers search spreadsheets by hand, long queues form, and a forwarded screenshot can be reused by someone else.
- ✔ **Approval.** Event plans go to the Youth Union / Student Association by e-mail or on paper; nobody can see the status of a request or the history of decisions.
- ✔ **Attendance evidence.** Conduct score depends on participation, but attendance is collected on sign-in sheets and re-typed later, so records are incomplete and appeals are hard to verify.

The main challenge is not the absence of individual tools for registration, payment, approval, or attendance tracking, but the **fragmentation** of these activities across disconnected systems. Student organisations must manually transfer and reconcile information between tools, creating

additional administrative effort, potential errors, and limited visibility into the overall event lifecycle.

UniEvent aims to consolidate event approval, registration, free and paid ticket management, automated VietQR payment confirmation and QR check-in into one platform.

Evidence: The project idea is informed by the first-hand experience of our team lead, Lê Nguyễn Quốc Học, a member of the Student Volunteer Team (Đội Sinh viên tình nguyện) who has helped organise several of its activities and attended many other university events. From these experiences, the team has observed workflows built on online registration forms, manually maintained participant lists, and attendance verification, and Học confirms that this is how such activities are run today. We treat these observations as preliminary domain knowledge, not as statistical evidence representing all student organisations. To avoid relying on a single organisation, we will compare this workflow with public event posts of other clubs (planned as a work package in PA1) and validate key assumptions with additional student organisers during the requirements phase (PA2). We also plan to ask the Student Volunteer Team to pilot UniEvent.

2.3 Target Users

User group	What they need from the system
Students (attendees)	Find events, buy tickets quickly with VietQR, keep tickets on the phone, see their attendance and conduct score history.
Club organisers	Create events, configure ticket types, follow sales and check-in in real time, export attendee lists, avoid manual payment reconciliation.
Check-in volunteers	Scan QR tickets on a phone and know instantly whether a ticket is valid or already used.
Youth Union / Student Association officers	Approve or reject event plans with comments, and review conduct score claims in bulk based on reliable attendance data.
System administrators	Manage clubs, accounts, roles and system-wide policies.

2.4 Main Features

Features F01-F08 are Must-have and will be fully implemented by PA5. F09 and F10 are Should-have; they will be specified and implemented if time allows.

#	Feature	Short description	Must-have?
1	F01 - Accounts and role-based access	Sign in with a university e-mail address; five roles with role-based access.	Yes
2	F02 - Event creation and submission	Organisers create events and submit them for approval.	Yes
3	F03 - Event approval workflow	Authorised officers review submitted events, approve, reject or request changes, and maintain a decision history.	Yes
4	F04 - Ticket type configuration	Price (free or paid), quantity, sales window, per-person limit.	Yes
5	F05 - Ticket reservation and VietQR payment	Paid tickets are temporarily reserved for 15 minutes pending payment via payOS. Free tickets bypass the payment process.	Yes
6	F06 - Payment confirmation and ticket issuing	The system verifies successful payments, prevents duplicate ticket issuance and generates a unique QR ticket for each confirmed registration. Tickets are delivered by e-mail.	Yes
7	F07 - QR check-in	Mobile-browser scanner; rejects used or invalid tickets instantly.	Yes
8	F08 - Real-time organiser dashboard	Live sales and check-in figures for organisers; CSV export.	Yes
9	F09 - Conduct score records	Students see their ticket and check-in status, with optional conduct score history.	No (Should)
10	F10 - Cancellation and refund policy	Ticket cancellation and refund decisions according to a policy.	No (Should)

2.5 Similar Products

✔ **Google Forms + Google Sheets + bank transfer** - A common way of working for student clubs, in our experience. Free and familiar, but it has no payment confirmation, no tickets, no check-in and no approval workflow. UniEvent automates exactly these steps.

✔ **Ticketbox** - A Vietnamese ticketing platform for public events such as concerts and festivals. UniEvent is simpler and campus-specific: free and low-cost tickets, university accounts, approval by the Youth Union and conduct score records, which a public ticketing platform does not provide.

- ✔ **Luma** – An international event-page and registration tool. It focuses on global payment methods, while UniEvent uses VietQR bank transfers, which Vietnamese students already use every day, and adds the university approval and conduct score workflow.

2.6 Platform and Technology Ideas

- ✔ **Platform:** responsive web application (desktop and mobile browsers); the check-in scanner runs in the mobile browser using the phone camera. No native app.
- ✔ **Front end:** Next.js (React, TypeScript) with Tailwind CSS.
- ✔ **Back end:** NestJS (TypeScript) on Node.js LTS, REST API and WebSocket gateway for real-time updates.
- ✔ **Database:** PostgreSQL with Prisma ORM.
- ✔ **External services:** payOS (VietQR payment links and webhooks), a transactional e-mail service.
- ✔ **Hosting:** Docker images; Vercel for the web app, Render or Railway for the API, Neon for PostgreSQL (free tiers).
- ✔ **Testing and CI:** Jest and Supertest, Playwright for end-to-end tests, k6 for load tests, GitHub Actions, ESLint and SonarQube Cloud.

These choices are not final; they will be confirmed with reasons in PA1.

2.7 Feasibility, Scope and Open Questions

Feasibility

We believe our team can build the eight Must-have features in weeks 8–10:

- ✔ The Must-have scope is split into five modules (accounts, events and approval, ticketing and payment, check-in, dashboard). Each member owns at least one module, so work proceeds in parallel.
- ✔ Most features are standard web functionality. The two technically risky parts - the payOS webhook and preventing overselling under concurrent purchases - will be tried in a technical spike in week 4, long before coding starts.
- ✔ Our estimate for the Must-have implementation is about 43 person-hours plus unit tests and reviews, against a capacity of about 144 person-hours in weeks 8-10 (4 members x 12 hours x 3 weeks, see Section 3.1). The remaining capacity covers testing, deployment and Should-have features.
- ✔ We will use AI coding assistants with full disclosure in the AI Usage Log; every member reviews and must be able to explain the code.

Out of scope

- ✔ Holding money on behalf of organisers (escrow) or paying out to clubs - this requires a payment-intermediary licence; payments go directly to the organiser's own payOS account.
- ✔ International card payments and currencies other than VND.
- ✔ Native iOS/Android applications - UniEvent is a responsive web application only.
- ✔ Direct integration with the university's official conduct score or student-information systems (UniEvent exports records instead).
- ✔ Seat maps and reserved seating.

Main risks

- ✔ payOS account or webhook problems – mitigated by an early spike and a mock payment adapter.
- ✔ Requirements biased towards one organisation – mitigated by comparing with other clubs' public event posts and keeping rules configurable.
- ✔ Members unable to explain AI-generated code – mitigated by mandatory reviews and weekly code walkthroughs.
- ✔ Under-estimated documentation effort – mitigated by internal deadlines two days before each PA.
- ✔ Free hosting being unavailable during the demo – mitigated by health checks and a backup video.

Questions for the TA

- ✔ May we integrate real payOS payments with small amounts for the demo, or should we use a mock payment adapter only?
- ✔ Should the PA documents be written in English or Vietnamese?
- ✔ Is there a minimum unit-test coverage that you expect in PA5?
- ✔ Our communication channel is the Discord server "UniEvent – T07", as required in PA1. Could you share your Discord username so that we can invite you as a member?

3 Team Contract

The team discussed each item below and agreed to follow it. The contract is reviewed when a conflict occurs and may be used as evidence when the teaching staff adjust individual contribution factors.

3.1 Goals and Expectations

- ✔ **Target grade:** Excellent level (90–100%) in every PA, the presentation and teamwork.
- ✔ **Effort:** Each member spends 8–10 hours per week on the project, and about 12 hours per week during the coding weeks 8–10.
- ✔ **Learning goals:** Practise the full waterfall life cycle, professional Git workflow (pull requests and reviews), automated testing and cloud deployment.

3.2 Communication

- ✔ **Main channel for discussion:** Discord server "UniEvent – T07" (https://discord.gg/zVMjhEGnzp); the TA is invited as a member.
- ✔ **Expected response time:** Within 12 hours on weekdays and 24 hours at weekends; urgent issues are tagged @everyone.
- ✔ **Where decisions and files are kept:** tasks on the Notion board "UniEvent Tasks" (approved by the TA); documents, diagrams and weekly reports in the GitHub repository (https://github.com/UniEvent-T07/UniEvent); architecture decisions as ADRs in /docs/design; decisions from meetings in the weekly status report.

3.3 Meetings

- ✔ **Weekly status meeting with the TA:** Thursday 20:00–20:15, 15 minutes, online.
- ✔ **Internal team meetings:** Tuesday 21:00–21:45 (planning, online) and Saturday 21:00–21:30 (progress and review, online); extra meetings before each PA deadline.
- ✔ **Attendance rule:** Inform the team at least 12 hours in advance if you cannot attend; absences are recorded in the weekly status report.
- ✔ **Weekly status report deadline:** Sunday 22:00 (GMT+7) for the week just finished, committed to /docs/management/weekly/ as W2-status.md, W3-status.md, … W10-status.md.

3.4 Roles and Responsibilities

Main and backup roles are listed in the team table in Section 1. Extra responsibilities:

Responsibility	Owner	Backup
Task board, weekly status report, Moodle submissions, TA contact	A	B
Requirements, traceability matrices, change log	B	A
Architecture, diagrams, consistency of IDs across PAs	C	A
Repository setup, branch protection, CI/CD and deployment	D	C
Test plan, test execution, defect log	D	B
AI usage log consolidation for each PA	B	D
Presentation slides and rehearsal	A	C
Meeting minutes	Rotating weekly	–

3.5 Work Standards

- ✔ **Internal deadline:** Each member finishes their part 2 days before a PA deadline; the whole document is reviewed and merged 1 day before the deadline.

- ✔ **Quality:** Every document section and pull request is reviewed by at least one other member before it is merged.
- ✔ **Tools:** Every task is on the task board with an owner and a due date; code and documents go through pull requests; no direct pushes to main.
- ✔ **Commits:** Small, frequent commits following Conventional Commits (feat:, fix:, docs:, test:, refactor:, chore:).
- ✔ **Definition of done:** The work meets its acceptance criteria, is reviewed, passes CI, and the related documents and traceability IDs are updated.

3.6 Use of AI Tools

- ✔ **Tools:** Claude, ChatGPT, Codex and GitHub Copilot (only the tools the team actually uses; any new tool is added to this list and to the AI Usage Log before use).
- ✔ **Allowed uses:** Brainstorming, drafting document sections, generating code and tests, explaining errors and reviewing drafts.
- ✔ **Review rule:** Every AI output is checked by its owner and by one other member; the owner must be able to explain every line that is submitted.
- ✔ **Logging:** Each member records significant AI use in /docs/ai-usage/PAx.md (tool, purpose, how the output was verified and changed); Member B consolidates the log into each PA.
- ✔ **Data protection:** No personal data of other people (e.g., interviewees, event participants), no real payment data and no secrets (API keys, passwords) are entered into AI tools; we use anonymised or synthetic data. Team members' own names and student IDs appear only in course documents.

3.7 Decision Making

- ✔ Decisions are made by consensus during team meetings or in the #decisions channel of our Discord server.
- ✔ If there is no consensus within 24 hours, the team votes; the majority decides.
- ✔ A tie is broken by the team lead; for architecture questions the architect (Member C) proposes options and the team votes.
- ✔ Important decisions are recorded in the weekly status report; architecture decisions are recorded as ADRs.

3.8 Handling Conflicts and Missed Work

- ✔ **Step 1:** talk directly with the member and agree on a fix and a new date.
- ✔ **Step 2:** if the new date is missed, discuss it in a team meeting, reassign the task if necessary and record it in the weekly status report.
- ✔ **Step 3:** if the problem continues (two or more missed tasks), inform the TA; the issue is taken into account in the individual contribution factor.
- ✔ A task that is 48 hours late without notice may be reassigned to its backup owner.

3.9 Signatures

By signing, each member agrees to follow this contract.

Student ID	Full name	Signature	Date
24127044	Lê Nguyễn Quốc Học	***Học***	12/10/2026
24127177	Thái Quang Huy	***Huy***	12/10/2026
24127196	Trần Vũ Anh Kiệt	***Kiệt***	12/10/2026
24127205	Nguyễn Văn Minh	***Minh***	12/10/2026

4 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Thái Quang Huy	Claude (Anthropic)	Drafted the project idea overview and the team contract for PA0 from the team's chosen idea.	All four members read the draft and discussed it in the team meeting on 10/10/2026. Changes made by the team: rewrote the Evidence paragraph to reflect Học's real experience; added the paragraph on tool fragmentation (2.2); chose Messenger as the communication channel and Saturday as the second internal meeting (1, 3.2, 3.3, 3.7); added a question for the TA about Messenger (2.7); filled in the team ID and the TA meeting time. On 12/10/2026 the team made further edits: softened general claims in 2.2 and 2.5 ("in our experience"), replaced a forward reference to the WBS in 2.2, removed the Luma domain name in 2.5, and corrected the AI data-protection wording in 3.6. Every member has read the final document and can explain it.
2	Lê Nguyễn Quốc Học	ChatGPT (OpenAI)	Reviewed the PA0 document against the official template and project guidelines to identify missing requirements, scope inconsistencies, and areas for improvement.	Manually reviewed the AI feedback and selected relevant suggestions. Clarified the proposed event approval model, reviewed the consistency between Must-have features and the product description, and refined the question for the TA regarding communication channels. Other suggestions were retained for consideration in PA1.

#	Member	Tool	Task / purpose	How the output was verified and changed
3	Thái Quang Huy	Claude (Anthropic)	Updated the communication channel from Messenger to Discord after the team decided to move (Sections 1, 2.7, 3.2, 3.7), and set up the Discord server structure and the Notion-to-Discord task notification. After the TA approved Notion as the task board, also updated Section 3.2 (task board) and added the Discord invite link in Section 1. On 13/10/2026, also proofread the final version (Sections 2.4 and 2.5) and added the repository link in Section 3.2.	Huy checked every changed sentence against the PA1 tool requirements (Slack, MS Teams or Discord, with the TA as a member). The changes will be reviewed by all members at the next team meeting. Entry 1 is kept unchanged as the record of the original decision.
4	Nguyễn Văn Minh	ChatGPT (OpenAI)	Reviewed the UniEvent project proposal to assess feature relevance, technical feasibility, project scope, and potential risks. Suggested improvements to feature prioritization and risk mitigation strategies.	AI feedback was compared with the PA0 proposal and course guidelines. Key suggestions concerning payment integration, QR check-in, feature scope, and development time were identified for team discussion. Only validated and agreed-upon recommendations will be incorporated into the project.